

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for economic control with recursive utility, endogenous preferences, and strategic interaction. Policy evaluation, feasible improvement, and verification are separate maps. A policy-specific continuous-time certificate connects a trained capital policy to the original economic optimum through a paired payoff identity, an explicit diffusion-transfer bound, and finite-family statistical inference. The deployed controller is specified by its observed innovations and sampled implementation; the economic state is not discretized or truncated in the guarantee. Unlike an architectural tube bound, the new endpoint depends on the fitted policy and can become tighter than the analytical schedule's bound. A rollout-costate implementation trains generic actors and critics in dense nonquadratic economies of dimensions ten, twenty, and fifty. Ten-seed comparisons include direct policy optimization, affine policies, tuned critic greedification, and accuracy-cost frontiers. The remaining economic applications retain their separate verification accounts. The results distinguish learned economic improvement, numerical safety, and method comparison without presuming optimizer convergence or universal neural superiority.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

Quantitative economic models increasingly place several margins of adjustment inside the same dynamic decision problem. An investor chooses consumption and risk exposure while managing a preference state. A sophisticated consumer chooses today knowing that tomorrow's self will reconsider. A firm invests while anticipating the investment policies of its rivals. These problems share a computational

structure: a continuation value must correctly evaluate a policy, and that policy must satisfy the appropriate optimality or equilibrium condition. The two requirements are distinct. A perfectly evaluated policy need not be optimal, and an attractive policy evaluated by an inaccurate continuation value need not improve welfare.

This paper develops Neural Bellman Operators (NBO) around this distinction. A critic represents continuation utility. A feasible actor represents the controls that enter the economic generator. The critic is fitted to a policy-evaluation equation, including its terminal and boundary data. Holding the critic and its derivative channels fixed, the actor improves its own Hamiltonian. An independent maximization step measures the remaining feasible-action gap. The resulting error decomposition concerns economic values and deviations, rather than only parameter gradients or the scalar quantity minimized by an optimizer.

The motivation remains continuous-time economic control, including recursive utility and costly preference formation in the motivating model of [Qi \(2025\)](#). We do not replace these problems by a prediction task. Instead, we make explicit how a neural approximation is connected to the original decision problem. In the smooth case, the connection follows from a verification bound. In the nonsmooth case, it follows from a monotone Bellman approximation and a comparison principle. The actor and critic may be trained by different numerical procedures in either case. Their economic interpretation is determined by the equations they satisfy and by the errors that are verified after training.

The paper makes four connected contributions. First, it specifies a neural policy-evaluation and improvement map with explicit derivative routing, boundary conditions, and an indexed evaluation guard. The guard bounds a selected-policy target residual by construction; it does not claim convergence of Adam or an uncorrected network. Second, it makes the complete-action error account operational through signed interval derivatives and verified face domination, with natural union bounds at stopping switches. A matched enclosure experiment isolates this calculation from policy training. Third, it carries the accounts through the original preference, capital, and strategic applications: nested-grid retraining, a model-specific compensating transfer, an absolute continuous-time capital benchmark, and finite dynamic best responses. Comparison, monotonicity, interval inclusion, and approximate policy

iteration remain inherited principles. The new results concern the specified guarded map, its verified output, and its measured economic accuracy, not an unqualified claim that a neural actor dominates classical or actor-free alternatives.

Fourth, an independently verified paired improvement tightens the continuous-capital policy bound after training. The evaluation block uses current-policy rollout and costate targets, while verification refers to a separately specified innovation-driven sampled controller. A new transfer estimate does not require a global derivative bound for the learned policy. The final experiment measures improvement relative to the analytical schedule with ten training seeds, explicit time-discretization and sampling uncertainty, and stronger tuned baselines. It preserves all earlier adverse evidence.

The numerical program uses generic multilayer critics and actors as well as clearly separated analytical laboratories. The finite preference and game studies train actors using exhaustively constructed continuation-payoff labels; their results establish neural policy representation, not avoided maximization. The continuous-action preference study instead differentiates the actor's own payoff, records local improvement as a hybrid policy, and reserves global action covering for independent verification after training. Direct optimization and an actor-free search ablation expose which parts of the computation produce the measured policy quality. Additional gated and projection comparisons exploit the same economic problem rather than weakening the baseline. Dense nonlinear capital experiments use paired pathwise payoffs to diagnose state coverage. Homothetic portfolios, temporal-self formulas, the nonsmooth exit problem, and the exact quadratic family remain distinct economic and algebraic checks. No matrix calculation is counted as neural training, and no grid refinement difference is silently converted into a continuous-time bound.

1.1. *Relation to the literature*

Neural residual methods build on numerical dynamic programming and function approximation, not on a new principle of optimality. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#).

Derivative approximation by smooth neural networks provides useful representation results ([Hornik et al., 1990](#)), but does not establish that a particular finite architecture or optimizer solves a nonlinear boundary-value problem.

The differential critic is closely related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. The distinguishing object here is the combination of an explicit feasible actor, block-specific evaluation, and economic error accounting across control and equilibrium problems. An analytical formula for the maximizing action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion ([Epstein and Zin, 1989](#), [Duffie and Epstein, 1992](#)). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; [Herdegen et al. \(2021\)](#) explains why a stationary algebraic solution alone is insufficient. The time-inconsistency extension is an equilibrium problem in the sense of [Björk et al. \(2016\)](#), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions ([Borkar, 2024](#)). They do not convert constant-step Adam into a convergence theorem for economic optimality.

The inherited continuous-capital result closes a different part of the error account: curvature and a verified coupling norm bound the gain from arbitrary adapted deviations, while a schedule-centred actor supplies a uniformly constrained neural policy. The resulting bound concerns the original diffusion rather than a nodal surrogate. Its stronger structural benchmark is reported alongside every fitted policy.

Section 2 specifies the controlled economy, Section 3 defines the NBO map, and Section 4 establishes the general error accounts. The continuous-capital theorem supplies an analytical benchmark. Sections 7–9 complete the policy-specific verification, algorithm, and economic comparison. The supplement retains the full

recursive-utility, preference, temporal-self, game, and scalability applications, all previous proofs, and the new transfer and arithmetic proofs.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant*

$L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q\sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this revision uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient

verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T}|V(S_T)] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section 26.

3. NEURAL BELLMAN OPERATORS

The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_\xi(t, x) = G(x) + (T - t)N_\xi(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_\xi = G_B + d_B N_\xi$.

Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state-time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_\nu \left[-\partial_t v_\xi - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_\xi] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_\nu \mathcal{H}^{a_\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction changes the population problem. With one action, $f(v) = -v$, $T = 1$, and terminal utility one, the correct solution is $v^*(t) = e^{t-1}$. The perturbation $v_\epsilon = v^* - \epsilon(1 - t)$ preserves the terminal condition. The joint objective used in the reviewed draft satisfies

$$\Delta(L_A + L_C) = -\frac{\epsilon}{2} + \frac{7\epsilon^2}{3}, \quad \Delta L_C = \frac{7\epsilon^2}{3}. \quad (8)$$

Thus the revised critic objective has the correct root as its minimizer along this perturbation, whereas the joint objective does not. The same problem is visible in a stationary equation: $-H + \omega H^2$ prefers $H = 1/(2\omega)$, not $H = 0$. No choice of finite positive weight resolves this identification error.

Algorithm 1 NBO evaluation, improvement, and acceptance

Require: Model, admissible actions, utility domain, boundary data, training and validation domains.

1: Initialize a feasible actor and a boundary-compatible critic.

2: **for** each outer iteration **do**

3: Freeze actor outputs. Update only the critic using (6) or a positive-weight Bellman evaluation residual.

4: Freeze the critic and every derivative channel. Improve only the actor using (7) or a feasible action solver.

5: Evaluate boundary error, policy-evaluation error, and the independent feasible-action gap.

6: Record the initialization, updates, conditioning, unsuccessful attempts, and stopping rule.

7: **if** the declared economic error criterion is met **then**

8: Return the policy, value, verification domain, and error report.

9: **end if**

10: **end for**

11: Return the computed approximation and its measured errors; do not infer optimality from optimizer stationarity.

The actor objective is a local policy-improvement surrogate. Under differentiability of the policy value, the derivative of lifetime recursive utility from (t, x) instead involves the adjoint weight

$$D_\phi V^{a_\phi}(t, x) = \mathbb{E}_{t,x} \int_t^\tau e^{\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv} D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds, \quad (9)$$

subject to the differentiability and integrability conditions given in the supplement. Arbitrary domain sampling does not reproduce this occupation and recursive-utility weighting. NBO uses (7) for approximate pointwise improvement and verifies the remaining action gap; it does not identify that loss with an exact lifetime policy gradient.

3.3. Conditioning and stopping

A positive, parameter-independent state weighting can precondition the critic equation without changing its zero set. If that weight depends on learned quantities, its derivative must be included in the stated critic objective. The homothetic laboratory divides the Hamiltonian by its positive coefficient and differentiates that entire normalized critic residual. In the actor step the same coefficient is detached. Error conversion to the original economic equation includes the scale factor.

A zero residual target does not make learning dynamics stationary. For the scalar loss $\ell(z) = a^2 z^2 / 2$, gradient descent is stable only when $0 < \eta a^2 < 2$. Likewise, full differentiation of $\delta^2 / 2$ gives $\delta \nabla \delta$, while a semi-gradient TD update differentiates only the current-value term. The two computations must not be compared using inconsistent signs or discount conventions. If f already includes $-\rho v$, a time discretization must not add a second independent discount to the same continuation equation.

Constant-step Adam is a numerical choice in the calibration laboratory, not an application of a two-timescale theorem. A genuine two-timescale analysis would require nonsummable and square-summable step sizes, an actor-to-critic ratio tending to zero, controlled noise, bounded iterates, and suitable attracting limiting dynamics (Borkar, 2024). A stationary critic gradient need not imply a zero residual, and a stationary multi-agent gradient need not imply Nash optimality. The results below therefore concern verified approximation errors rather than an assumed universal optimizer convergence property.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections 5 and 26.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

3.5. Continuous-action proposals and a budget-safe Bellman enclosure

The learned map and its verifier have different computational roles. In the differential implementation, a feasible actor supplies an action and the critic evaluates that action’s generator. In the positive-weight implementation, write

$$(T_n^a w)_i = r_{n,i}(a) + q_n \{P_{n,i}(a)w + b_{n,i}(a)\}, \quad (T_n w)_i = \sup_{a \in \mathcal{A}_i} (T_n^a w)_i. \quad (11)$$

Here $P_{n,i}(a)$ has nonnegative entries with sum at most one. The term b contains prescribed stopping payoffs, and $q_n > 0$. Continuous controls need not enter this expression concavely. Reflection, interpolation cells, and stopping decisions can change as the action changes. A supremum, rather than an attained maximizer, suffices below.

Our continuous-action implementation fits the actor by differentiating its own evaluated continuation payoff. It does not construct an all-action target or receive an exact maximizing label. A bounded tanh output enforces the control box. A

local improvement step can subsequently compare that proposal with finitely many continuous perturbations. Such a step is recorded as part of a *hybrid* policy; it is not identified with the raw network and is not assumed globally optimal. Critic fitting uses the selected feasible action and the preceding continuation critic. Training ends before any exhaustive reference or action-cover verification begins.

For independent verification, freeze the complete policy $\pi = (\pi_n)$ and cover each action box by closed subboxes. At a state and time, let c_i be an outward lower bound obtained by evaluating any feasible action using a fixed continuation upper array. A box can be discarded only when its verified upper bound does not exceed $c_i + \tau$. Every box not so discarded is either subdivided or retained with its upper bound. Exhausting a time, depth, or box budget therefore enlarges the reported bound; it never converts an unresolved search into a success.

PROPOSITION 1—Budget-safe action and policy envelopes: *Suppose the terminal arrays satisfy $L_N \leq g \leq U_N$. At each time n , evaluate the fixed-policy map outward to obtain $L_n \leq T_n^{\pi_n} L_{n+1}$. For every state, maintain a complete action cover whose box bounds enclose $T_n^a U_{n+1}$ for all actions in that box. Set $U_{n,i}$ to the maximum of the outward-rounded pruning threshold and all unresolved box upper bounds. Then, regardless of which boxes remain unresolved,*

$$L_n \leq V_n^\pi \leq V_n^* \leq U_n, \quad 0 \leq V_n^* - V_n^\pi \leq U_n - L_n. \quad (12)$$

All inequalities are on the stated nodal economy. No concavity, exact training labels, or convergence of neural optimization is required. If $U_n - T_n U_{n+1} \leq e_n \mathbf{1}$, then its optimal-value excess is at most

$$\left(\prod_{j=n}^{N-1} q_j \right) \|U_N - g\|_\infty + \sum_{k=n}^{N-1} \left(\prod_{j=n}^{k-1} q_j \right) e_k.$$

The proof, including stopping faces, is in the supplement. A stored unresolved bracket supplies an upper bound for e_n ; it is not replaced by the requested tolerance. The policy lower envelope and the Bellman upper envelope are computed separately. The latter may be shared across several frozen policies on the same economy, while

each policy requires its own evaluation. This separation prevents an expensive verification oracle from becoming undisclosed actor supervision.

In the preference application, the continuous box has three coordinates and the state grid has two. Bilinear interpolation is enclosed cell by cell. On each rectangle, its extrema occur at cell corners; a second implementation uses cell-slope enclosures. If a box crosses a wealth stopping face, the verifier takes the union of the interior continuation and the prescribed exit payoff. Their numerical values need not coincide at off-grid exit locations. Treating the entire action objective as globally differentiable at this switch would invalidate a derivative-only bound. Reflection is handled by including every crossed fold. Positive consumption and a preference interval strictly above one keep the powers and denominators in their stated domains.

The complete-action bound is not a continuous-state or continuous-time theorem. The verifier interprets stored primitive coefficients and nodal abscissae as exact real inputs and encloses subsequent operations outward. Training and verification agree to numerical precision on the inherited action grid, but the certificate’s precise object is this declared nodal economy, not an unspecified sequence of floating-point evaluations. The supplemental transfer result states separately the state interpolation, diffusion approximation, boundary monitoring, and action errors needed to recover the continuous economy. None is inferred from a sampled residual or silently assigned zero.

4. VERIFICATION, APPROXIMATION, AND VISCOSITY SELECTION

For a feasible policy a and a smooth candidate v , define

$$r^a[v] = -v_t - \mathcal{H}^a[v], \quad g^a[v] = \sup_{b \in \mathcal{A}} \mathcal{H}^b[v] - \mathcal{H}^a[v] \geq 0. \quad (13)$$

The first error concerns evaluation; the second concerns improvement. Their difference is the maximized HJB residual: $F[v] = r^a[v] - g^a[v]$. We keep the two quantities separate because the policy value and its regret depend differently on them.

THEOREM 2—Finite-horizon economic error bound: *Under Assumption 1, suppose a bounded smooth candidate v in the declared utility interval and an admissible Markov*

policy a satisfy $|r^a[v]| \leq e$, $0 \leq g^a[v] \leq \eta$ throughout the verification domain. Suppose the boundary payoff error is at most B . Any reflected boundary condition is imposed exactly. Set $A_L(s) = (e^{Ls} - 1)/L$ for $L > 0$ and $A_0(s) = s$. Then, for $s = T - t$,

$$|v(t, x) - V^a(t, x)| \leq e^{Ls}B + A_L(s)e, \quad (14)$$

$$|v(t, x) - V^*(t, x)| \leq e^{Ls}B + A_L(s)(e + \eta), \quad (15)$$

$$0 \leq V^*(t, x) - V^a(t, x) \leq 2e^{Ls}B + A_L(s)(2e + \eta). \quad (16)$$

Consequently, exact evaluation, exact boundary data, and a zero feasible-action gap identify the optimal value and an optimal policy.

The theorem replaces an assertion about a global minimizer of a composite loss with a statement about the exact computational objects NBO estimates. It allows errors, does not require that a finite network class contain V^* , and bounds performance against all admissible policies covered by comparison. A small sampled residual is not yet the uniform hypothesis of the theorem. Section 4.3 addresses that distinction.

4.1. A positive-weight recursive Bellman approximation

Let P_h^a be a nonnegative transition/interpolation operator on a finite state set, with row sums one after adjoining absorbing states. Boundary values are fixed. For a continuation array w , define $\Psi_h^a w$ pointwise as the scalar solution

$$z = P_h^a w + hf(x, a, z), \quad T_h w = \max_{a \in \mathcal{A}_h(x)} \Psi_h^a w. \quad (17)$$

For stationary calculations assume $f_y \leq -\lambda < 0$ on an invariant value domain, and require that all scalar roots remain in that domain. The map $z \mapsto z - hf(x, a, z)$ is strictly increasing with slope at least $1 + \lambda h$. Hence the root is unique and Ψ_h^a is order preserving and a sup-norm contraction with factor $q_h = (1 + \lambda h)^{-1}$. The maximum over feasible actions preserves these properties. A root bracket must contain the root; arbitrary clipping of utility values is not part of the definition.

For a finite horizon, the same implicit construction is used backward from the prescribed terminal array. Under the Lipschitz condition in Assumption 1, $hL < 1$

gives a one-step Lipschitz constant at most $(1 - hL)^{-1}$. Discount contraction is not assumed for a recursive parameter regime where it has not been verified. For a time-additive problem, the positive-weight backup $hr_0 + e^{-\rho h} P_h^a w$ is also consistent and monotone. It is the backup used by the NDU reference laboratory.

THEOREM 3—Discrete NBO certificate: *Suppose (17) is well defined on an invariant domain, with contraction factor $q_h < 1$. Let v_h^* and v_h^a denote the fixed points of T_h and T_h^a , respectively. For any neural value array v and deterministic feasible policy a , set*

$$\delta = \|v - T_h^a v\|_\infty, \quad \epsilon = \|T_h v - T_h^a v\|_\infty.$$

Then

$$\|v - v_h^a\|_\infty \leq \frac{\delta}{1 - q_h}, \quad \|v - v_h^*\|_\infty \leq \frac{\delta + \epsilon}{1 - q_h}, \quad \|v_h^* - v_h^a\|_\infty \leq \frac{2\delta + \epsilon}{1 - q_h}. \quad (18)$$

The certificate covers the stated finite state and action model. Continuous actions require an additional certified maximization or action-coverage error.

This result applies regardless of how the neural array was obtained. Neural projection need not itself be monotone. What matters is that the output satisfies the monotone Bellman equation to the displayed tolerance. The distinction makes it possible to use flexible representations while retaining a comparison-based acceptance test.

THEOREM 4—Viscosity-consistent neural Bellman approximation: *Consider stationary problems satisfying comparison in a bounded continuous class and the strict utility monotonicity used in Theorem 3. Suppose the positive-weight schemes are stable, locally consistent with the HJB operator, and attain the prescribed boundary data in the limit. The feasible action approximations become dense with vanishing maximization error. If neural arrays and feasible actors satisfy*

$$\delta_h + \epsilon_h = o(h), \quad (19)$$

then their interpolated values converge locally uniformly to the unique viscosity solution. Uniform convergence up to the boundary follows when boundary barriers give uniform

boundary attainment. Their discrete policy-value arrays converge to the same limit. No assertion of convergence of the policy functions themselves is required.

Here stability, local consistency, comparison, and boundary attainment are numerical-analysis hypotheses, not consequences of calling a representation neural. The proof is given in the appendix. A finite-horizon version uses backward error accumulation: terminal error tends to zero and the sum of one-step errors tends to zero after multiplication by the one-step Lipschitz factors. This covers finite-horizon recursive problems without imposing stationary discount contraction.

For a diffusion with m Brownian factors, an interior semi-Lagrangian construction uses the $2m$ locations $x + hb \pm \sqrt{mh} \sigma e_j$, each with weight $1/(2m)$. Positive multilinear interpolation preserves monotonicity. For smooth test functions its interpolation contribution is $O(\Delta x^2/h)$; thus $\Delta x^2/h \rightarrow 0$ is a sufficient interior coupling condition. The mean and covariance of the cubature increments match those of the specified Brownian diffusion to first order. Stopping and reflection require consistent boundary transitions. Negative interpolation weights are not permitted in this construction.

4.2. Why an integrated differential residual does not select the value

Consider the exit problem $F(v') = 1 - |v'| = 0$ on $(-1, 1)$ with zero boundary payoff. Its value is $v^*(x) = |x| - 1$. The smooth functions

$$w_\epsilon(x) = \sqrt{1 + \epsilon^2} - \sqrt{x^2 + \epsilon^2}$$

satisfy the same boundary values and have vanishing integrated squared residual, but converge uniformly to $1 - |x|$. At zero the test function one touches the wrong limit from above and gives $F(0) = 1 > 0$, violating the viscosity subsolution inequality. The defect is not removed by exact Hamiltonian maximization.

On a grid with spacing h , the positive-weight exit backup is

$$(Tv)_i = -h + \max(v_{i-1}, v_{i+1}), \quad v_0 = v_N = 0. \quad (20)$$

It is an undiscounted shortest-exit problem, not an instance of the stationary contraction theorem. Backward propagation of the boundary values gives $v_i = |x_i| - 1$

exactly. At the peak of the wrong candidate, the *scaled* residual $|w_i - (Tw)_i|/h$ is close to two, rather than close to zero. The laboratory therefore distinguishes correct selection from small integrated differential error.

An explicit positive viscosity term is another possible regularization, provided comparison, boundary convergence, and approximation errors for the regularized equations are controlled before taking the viscosity to zero. Network width or spectral bias is not itself a positive viscosity parameter. Moreover, approximation of a C^2 function and its derivatives does not establish approximation of nonexistent classical derivatives at a kink.

4.3. Coverage and independent maximization

A verification distribution is declared independently of the actor’s occupation measure. It includes economically relevant counterfactual states and boundary regions, not only states visited by a current policy. If a residual r has a known Lipschitz bound K on a compact domain and a validation set has fill distance d_{fill} , then

$$\|r\|_{\infty} \leq \max_{x \in \mathcal{V}} |r(x)| + Kd_{\text{fill}}. \quad (21)$$

The same bound can be applied to an action gap if its modulus is controlled. An empirical mean alone does not give this guarantee. If the sampling density is bounded below and the domain has a uniform interior-volume condition, a Lipschitz residual also admits an L^2 -to-sup estimate with exponent $1/(d+2)$, as derived in the supplement. Its dimension dependence is material.

An independent action solver must provide an *upper bound* on the unobserved maximization error. Trying a few alternative actions supplies only a lower bound on the true gap. Concavity with a duality gap, interval bounds, exhaustive enumeration of a finite action set, or an action net with a known Lipschitz modulus can provide the needed upper bound. An unverified local actor optimizer cannot certify itself.

4.4. Differential and discrete error accounts

For the stationary additive problem, write $f = r_0 - \rho v$, with $\rho > 0$, and let R_h restrict a candidate to the nodes of a positive-weight scheme. On interior nodes set

$$T_h^a w = q_h(P_h^a w + h r_0^a), \quad q_h = (1 + \rho h)^{-1}.$$

Stopping nodes instead use their prescribed payoff.

PROPOSITION 5—Transfer to a Bellman certificate: *Suppose the represented policy is feasible for the discrete action class and*

$$\sup_a \left\| h^{-1}(P_h^a R_h v - R_h v) - R_h \mathcal{L}^a v \right\|_{\infty, \text{int}} \leq \kappa_h.$$

If the differential evaluation residual is at most e and its feasible-action gap at most η , then, on the interior nodes,

$$\|R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(e + \kappa_h), \quad \|T_h R_h v - T_h^a R_h v\|_{\infty} \leq q_h h(\eta + 2\kappa_h).$$

Boundary discrepancies and finite-action coverage errors are additional accounts.

The proof follows by adding and subtracting $R_h \mathcal{L}^a v$; the maximization inequality uses the consistency error twice. The converse is false for arbitrary oscillatory interpolants. For smooth interior functions, local cubature and interpolation estimates can bound κ_h ; they do not justify assigning zero error to a stopping or reflected boundary without checking its realization.

For finite-horizon maps with Lipschitz factors β_t , let δ_t bound evaluation error, ϵ_t bound improvement error, and B_N bound terminal error. Define

$$E_N = B_N, \quad O_N = B_N, \quad R_N = 2B_N, \quad \begin{cases} E_t = \delta_t + \beta_t E_{t+1}, \\ O_t = \delta_t + \epsilon_t + \beta_t O_{t+1}, \\ R_t = 2\delta_t + \epsilon_t + \beta_t R_{t+1}. \end{cases} \quad (22)$$

Backward comparison bounds policy-value error by E_t , optimal-value error by O_t , and policy regret by R_t . These arrays are computed, rather than replaced by a training loss, in the NDU experiment. Bounds at later time indices must use

their own accounts, not necessarily the account at time zero. Representation and incomplete optimization enter δ_t ; action coverage enters ϵ_t ; the error of replacing the controlled diffusion by a scheme is separate. If a uniform continuous-to-discrete policy-payoff discrepancy is bounded by D_h , transferring a discrete regret bound adds $2D_h$. Without that estimate, the certificate remains a certificate for the stated discrete economy. In particular, this accounting does not establish the optimizer rate required by Theorem 4.

5. A TRAINED NEURAL BOUNDARY-VALUE PROBLEM

Consider a household with labor income, risky wealth, and a prescribed retirement payoff on exit. Before exit from $D = (0.5, 2.5)$,

$$dX_s = (0.12 + 0.02X_s - c_s) ds + 0.15X_s dW_s, \quad c_s \in [0.02, 0.6]. \quad (23)$$

Discount is $\rho = 0.08$, flow utility is $\log c$, and exit pays $g(x) = \log(0.12 + 0.08x)/0.08$. This is a specified stopped economy, not a claim about an unbounded-wealth problem. Income and the exit option destroy the homothetic reduction used in the portfolio calibration. Bounded coefficients, a compact positive consumption interval, positive diffusion variance on the closed domain, and continuous endpoint data yield the usual stopped-domain comparison problem. A neural approximation must solve its nonlinear differential equation and both boundary conditions.

The critic has two hidden tanh layers of width sixteen and the lifting

$$v_{\xi}(x) = g(\ell) + \frac{g(u) - g(\ell)}{u - \ell}(x - \ell) + (x - \ell)(u - x)N_{\xi}\left(2\frac{x - \ell}{u - \ell} - 1\right).$$

The actor uses the same hidden architecture and a sigmoid into the consumption interval. Here the optimum is not forced to an endpoint by that representation. Independent verification nevertheless maximizes over the *closed* interval: for $p = v'(x)$,

$$c^*(p) = \begin{cases} \text{clip}(1/p, 0.02, 0.6), & p > 0, \\ 0.6, & p \leq 0. \end{cases} \quad (24)$$

Training imports the general differential core used for the earlier routing tests. It does not fit reference values, derivatives, or policy labels.

PROPOSITION 6—Continuous-domain economic certificate: *For a stopped dis-*
counted additive economy obeying comparison, let $v \in C^2(D) \cap C(\overline{D})$ and let a be admissi-
ble. Suppose throughout D that

$$|\rho v - r_0^a - \mathcal{L}^a v| \leq e, \quad 0 \leq \sup_b (r_0^b + \mathcal{L}^b v) - (r_0^a + \mathcal{L}^a v) \leq \eta, \quad \|v - g\|_{\partial D} \leq B.$$

Then

$$\|v - V^a\|_\infty \leq B + \frac{e}{\rho}, \quad \|v - V^*\|_\infty \leq B + \frac{e + \eta}{\rho}, \quad 0 \leq V^* - V^a \leq 2B + \frac{2e + \eta}{\rho}. \quad (25)$$

Indeed, constant upper and lower shifts of v dominate the evaluation or optimality residual and the stopping data; comparison gives the first two inequalities and their sum gives the third. No reference-grid error is used in this argument.

The executable verifier covers the entire state interval by dyadic subintervals. Outward-rounded propagation through the stored network encloses v, v', v'' , and the actor on every leaf. Equation (24) supplies a global action enclosure. We also use the concavity bound

$$\log c^* - pc^* - (\log c - pc) \leq \frac{1}{2}C^2(1/c - p)^2, \quad C \geq \max\{c, c^*\}, \quad (26)$$

which follows by completing the square in the strong-concavity inequality on the segment joining the two actions. It does not assume an interior maximizer. Taking the better of this bound and a direct interval enclosure avoids dependency inflation. Terminal-free boundary errors are evaluated separately at both endpoints, including rounding in the lifting constants.

The certificate saves every leaf, verifies that their union is exactly $[0.5, 2.5]$, and binds the parameter and arithmetic files by hashes. Exponential and logarithmic enclosures use range reduction and polynomial remainder bounds rather than an assumed accuracy of a library transcendental. Its mathematical conclusion is conditional on the stated binary64 arithmetic contract; high-precision and derivative tests check the implementation but are not a formal machine proof. The construction does not estimate a global Lipschitz constant from samples.

TABLE I
MULTILAYER CONSUMPTION: APPROXIMATION AND CONTINUOUS-DOMAIN VERIFICATION

Method	Seed	$\ v - V_h^*\ _\infty$	Sampled e	Uniform η	Regret bound
NBO	11	3.30×10^{-4}	6.18×10^{-4}	4.24×10^{-6}	0.0501
NBO	29	5.75×10^{-4}	0.0015	2.66×10^{-6}	0.0500
NBO	47	5.46×10^{-4}	0.0014	4.36×10^{-6}	0.0501
Direct HJB	11	4.14×10^{-4}	9.72×10^{-4}	0	0.0500
Direct HJB	29	3.55×10^{-4}	7.13×10^{-4}	0	0.0500
Direct HJB	47	6.00×10^{-4}	0.0017	0	0.0500
Joint loss	11	5.4142	0.5002	—	—
Joint loss	29	5.4435	0.5003	—	—
Joint loss	47	5.5677	0.5003	—	—

The comparison grid has 2,560 intervals on $[.5, 2.5]$. The last two columns cover the entire continuous interval and consumption set $[.02, .6]$. All six certificates have uniform evaluation error at most .002 and endpoint error below 10^{-12} . Dashes indicate no accepted certificate, not zero error. Reference error and the sampled differential residual are not used as uniform bounds.

5.1. Controlled comparison and failures

We use seeds 11, 29, and 47, each with 256 prespecified jittered collocation points. Eight outer rounds have a common cap of 800 L-BFGS inner iterations. NBO allocates each round sixty iterations to evaluation and forty to improvement; direct neural HJB uses the same critic and the global action rule in all one hundred iterations. A joint-loss ablation differentiates the sum of squared residual and negative Hamiltonian through both blocks. Line searches use different numbers of closures, so this is a matched iteration cap, not an assertion of equal floating-point work. Closure counts and elapsed times are saved.

An independent positive-rate upwind policy-iteration solver uses continuous-action maximization within each drift region. Reference grids have 320, 640, 1,280, and 2,560 intervals. Value and derivative comparisons use the finest grid; its refinement differences are not substituted for a rigorous discretization bound. The continuous certificate instead checks the neural functions directly. All three NBO and direct-HJB runs attain the declared interval-verification thresholds. The joint-

loss ablation does not: a residual near one half is accompanied by large value and action errors. Width, depth, and update-allocation variations are retained in the supplement, including unfavorable outcomes. The direct method is a strong comparator: these results do not establish that carrying an actor is uniformly faster or more accurate. They establish an end-to-end neural solve with an independently computable welfare bound, and identify a concrete failure of joint differentiation. Fitting and full-domain verification are separately timed because their computational burdens are different.

6. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule. Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z} \mathbf{1}$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - \nu^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (27)$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{\overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2\} dt + e^{-\rho T} \{\bar{Y}_T - \chi \|\Pi Y_T\|_d^2\} \right], \quad (28)$$

where $\nu^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1$, $\rho = .04$, $a = .10$, $\kappa = .15$, $\sigma_I = .15$, $\sigma_C = .10$, $\zeta = .2$, $\chi = .03$, and $[\underline{m}, M] = [.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (29)$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. Let $\beta \geq \kappa \|B\|_2$, let s_0 bound the initial cross-sectional standard deviation, and set

$$Q_d = s_0 + \kappa T + \sigma_I \sqrt{(1 - 1/d)T}, \quad (30)$$

$$K(s) = \beta \int_s^T e^{-\rho t} w(t) e^{\beta(t-s)} dt + 2\chi e^{-\rho T} Q_d e^{\beta(T-s)}, \quad (31)$$

$$\mu(t) = \frac{2\{\log[m_0(t)/M] + [M - m_0(t)]/m_0(t)\}}{[M - m_0(t)]^2}. \quad (32)$$

The quantity $\mu(t)$ is the smallest tangent-remainder curvature of log utility on the entire primitive action interval, relative to $m_0(t)$. It is not a Hessian bound estimated from training points. In particular, it can be substantially sharper than the uniform lower curvature M^{-2} .

THEOREM 7—Curvature-controlled policy guarantee: Suppose $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, the schedule in (29) is interior, and $\|\Pi y\|_d \leq s_0$. Let $V^*(y) = \sup_m J_y(m)$, where the supremum is over the full original admissible class. Then

$$0 \leq V^*(y) - J_y(m_0) \leq G_d := \frac{1}{2} \int_0^T e^{\rho s} \frac{K(s)^2}{\mu(s)} ds. \quad (33)$$

For $\epsilon \geq 0$ satisfying $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$, every admissible policy m^ϕ with $|m_i^\phi(t, Y) - m_0(t)| \leq \epsilon$ for all coordinates, times, and states satisfies

$$0 \leq V^*(y) - J_y(m^\phi) \leq G_d + D_{d,\epsilon}, \quad (34)$$

where, with $A_\beta(T) = \int_0^T e^{\beta t} dt$,

$$D_{d,\epsilon} = \int_0^T \left[\epsilon K(s) + \frac{\epsilon^2 e^{-\rho s}}{2} \{[m_0(s) - \epsilon]^{-2} + \zeta\} \right] ds + \chi e^{-\rho T} \epsilon^2 A_\beta(T)^2. \quad (35)$$

These bounds hold for every initial mean, every initial state with the stated dispersion bound, and arbitrary adapted competitors in the original action box. They require no neural training convergence or discretization error estimate.

The proof uses synchronous state comparison, an exact discounted mean-state identity, and completion of a quadratic bound on an arbitrary policy deviation. It is provided in the appendix. The result is a model-specific numerical principle: curvature limits the benefit from changing consumption, whereas a spectral bound limits the propagation of that change through dense nonlinear production. Their balance yields a scalar integral certificate for a high-dimensional policy. Bounded nonlinearity, the additive diffusion structure, and the chosen felicity are essential assumptions, rather than generic properties of all HJB equations.

6.1. The deployed neural map and its numerical guard

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (36)$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (37)$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its interval midpoint and is not evidence of successful training. This scalar feasibility

guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

6.2. Computable scope and cost

An SVD is used only to propose a scalar λ . Positive pivots in an outward interval $LDL^\top$ calculation for $\lambda I - B^\top B$ certify the spectral majorant. A failed pivot triggers enlargement, not acceptance of the SVD output. The remaining integrals use outward rectangle sums, polynomial exponential and logarithm bounds, and square-root endpoints checked by interval squaring. For a fixed majorant with verified pivots, this certificate requires $O(d^3)$ arithmetic operations and $O(d^2)$ storage for the matrix check, followed by $O(N)$ scalar quadrature work. This is not a dimension-independent training theorem or a worst-case bound on the number of attempted majorants.

The guarantee holds already at initialization. Training is intended to improve the feasible policy inside the tube, not to manufacture its certificate. The anchor itself has the tighter bound G_d and requires no training. A comparison omitting it would overstate the role of the actor. Neither a small $G_d + D_{d,\epsilon}$ nor faster actor evaluation proves a net economic advantage over that benchmark.

7. POLICY-SPECIFIC VERIFICATION AFTER BELLMAN IMPROVEMENT

The bound in Theorem 7 is a useful starting point, but not a measure of what training accomplishes. Every policy in the same tube receives the same allowance, including a poorly trained policy. We now replace that allowance by an independently verified, policy-specific payoff difference. The economic primitives, initial state, and full adapted comparison class in (27)–(28) are unchanged. The implementation to which the new bound applies is specified explicitly below. It is not identified with continuously observed-state feedback.

7.1. The deployed controller

Fix a deterministic grid $t_k = kh$, $Nh = T$. The controller has an internal state Z_k , initially y , and observes the Brownian innovations at the grid dates. At date t_k it

selects

$$a_k = \mathcal{P}_k\{m_\phi(t_k, Z_k)\}, \quad Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\Delta\tilde{W}_k, \quad (38)$$

where $c = a - v^2/2$, $f(z) = \kappa \tanh(Bz)$, and $\mathcal{P}_k$ projects inward onto the declared action interval around $m_0(t_k)$. The vector $\Delta\tilde{W}_k$ clips each standardized innovation at ten before rescaling by $\sqrt{h}$. The numerical implementation and its rounding convention form part of the controller. Between dates it holds a_k fixed. Its economic state, denoted Y^ϕ , nevertheless satisfies

$$dY_t^\phi = \{c\mathbf{1} + f(Y_t^\phi) - a_k\} dt + \Sigma dW_t, \quad t_k \leq t < t_{k+1}.$$

In particular, the economic Brownian motion and state are not clipped, stopped, or replaced by Z . The controller is adapted to the original innovation filtration and remains feasible. Observing innovations is an implementation assumption, not a conclusion inferred from a state-only observation model. The continuously evaluated R10 policies remain separate objects in the supplement.

This specification has an important analytical consequence. When the numerical state and the economic state are compared, they use exactly the same chosen actions. Their state discrepancy therefore depends on the smoothness of the economic drift, not on a global Lipschitz bound for the learned network. The following construction uses this fact rather than inserting sampled residuals into a uniform verification theorem.

7.2. An exact improvement identity

Write $W(t) = e^{-\rho t}w(t)$ and define the nonnegative consumption deficit

$$R_t(m) = \log m_0(t) - \overline{\log m} + w(t)\{\bar{m} - m_0(t)\} + \frac{\zeta}{2}\{\bar{m}^2 - m_0(t)^2\}. \quad (39)$$

The first-order identity $m_0^{-1} = w + \zeta m_0$ and concavity of the logarithm imply $R_t(m) \geq 0$. The exact mean-state identity in the proof of Theorem 7 gives

$$\begin{aligned} \Delta_\phi := J_y(a^\phi) - J_y(m_0) &= \mathbb{E} \int_0^T W(t) \overline{f(Y_t^\phi) - f(Y_t^0)} dt - \mathbb{E} \int_0^T e^{-\rho t} R_t(a_t^\phi) dt \\ &\quad - \chi e^{-\rho T} \mathbb{E}\{\|\Pi Y_T^\phi\|_d^2 - \|\Pi Y_T^0\|_d^2\}. \end{aligned} \quad (40)$$

Thus state-dependent consumption has to earn its utility-curvature cost through production or terminal dispersion. The identity is also an effective variance reduction: the common mean-state martingale cancels before simulation statistics are formed. Positive neural output is not itself evidence of positive economic improvement.

For each cell, calculate the scalar integrals

$$\begin{aligned} A_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, & B_k &= \int_{t_k}^{t_{k+1}} W(t) dt, \\ C_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} \{\log m_0(t) - w(t)m_0(t) - \zeta m_0(t)^2/2\} dt. \end{aligned}$$

The consumption-deficit integral for a held action is exactly $C_k - A_k \overline{\log a_k} + B_k \bar{a}_k + \zeta A_k \bar{a}_k^2/2$. Consequently, consumption need not be approximated by a left-point utility sum. Let $X_{\phi,h}$ be the paired statistic obtained from (40) by using the internal Euler states in the production and terminal terms and these integrated consumption terms. The reference internal state uses the integrated common action $\int_{t_k}^{t_{k+1}} m_0(t) dt$. All scalar integral enclosures and their midpoint errors are saved.

7.3. A network-independent diffusion-transfer bound

Here the word independent concerns the transfer constants, not the economic statistic. Let

$$\begin{aligned} q_i &= \sigma_I^2 \|B_i\|_2^2 + \sigma_C^2 \left(\sum_j B_{ij} \right)^2, \\ H &= \kappa \left(d^{-1} \sum_i q_i \right)^{1/2}, & Q &= \frac{\kappa}{2} \left(d^{-1} \sum_i q_i^2 \right)^{1/2}. \end{aligned}$$

The letter Q here denotes a generator bound, not the dispersion bound Q_d in (30). For a policy in a radius- ϵ interval at sampling dates, put

$$M_\epsilon = \max_{m \in [m_0(0) - \epsilon, m_0(T) + \epsilon]} |c - m| + \kappa, \quad K_\epsilon = \beta M_\epsilon + Q.$$

The terminal dispersion majorant is

$$S_\epsilon = s_0 + (\kappa + \epsilon)T + \sigma_I \sqrt{(1 - 1/d)T}.$$

The formula uses the monotonicity of this economy's m_0 . For more general schedules their extrema replace the endpoints. Define

$$u_\epsilon(t) = he^{\beta t} \{K_\epsilon t/2 + H\sqrt{t/3}\}, \quad (41)$$

$$E_\epsilon(h) = \beta \sum_k B_k u_\epsilon(t_k) + K_\epsilon \sum_k \int_{t_k}^{t_{k+1}} W(t)(t - t_k) dt \\ + \chi e^{-\rho T} u_\epsilon(T) \{2S_\epsilon + u_\epsilon(T)\}. \quad (42)$$

PROPOSITION 8—Transfer of paired policy values: *For exact arithmetic and untruncated internal innovations, the preceding construction satisfies*

$$|\Delta_\phi - \mathbb{E}X_{\phi,h}| \leq E_\epsilon(h) + E_0(h).$$

With clipped internal innovations, forward rounding, and enclosed scalar integration, replace each u_ϵ by its stated enlarged bound and add the statistic's quadrature and arithmetic error. Denote the resulting saved upper bound by $E_{\phi,h}^{\text{all}}$. It is $O(h)$ apart from these explicit numerical terms and does not involve a network derivative bound. The economic process remains the original continuous diffusion.

The proof is in the supplement. Its key step is to retain the martingale cancellation in the accumulated local drift error. Bounding each local stochastic error separately by its absolute magnitude would yield only an $O(\sqrt{h})$ account and would obscure the small policy differences studied here. This proposition concerns the sampled implementation (38); it does not prove this rate for arbitrary state-observed neural feedback.

7.4. A post-training certificate

Paired statistics are not uniformly bounded, because the terminal dispersion difference contains Gaussian shocks. This is dealt with explicitly, rather than ignored in a bounded concentration inequality. For each declared initial state and radius, the supplement constructs constants C_ϕ, A_ϕ such that

$$\Pr\{|X_{\phi,h}| > C_\phi + A_\phi(\sqrt{d-1} + v)\} \leq e^{-v^2/2}, \quad v \geq 0.$$

They are obtained from synchronous state separation and the norm of the projected reference innovations. Neither an empirical output range nor an empirical maximum enters their construction. With $v_* = 8$, put $b_\phi = C_\phi + A_\phi(\sqrt{d-1} + v_*)$ and $\tau_\phi = A_\phi e^{-v_*^2/2}/v_*$. Then clipping $X_{\phi,h}$ to $[-b_\phi, b_\phi]$ changes its expectation by at most τ_ϕ .

For n fresh independent paths, let $\bar{X}_\phi^c$ and s_ϕ^2 be the clipped empirical mean and unbiased variance. Fix a family of at most F one-sided statements before inspecting this noise bank, and set $\ell = \log(2F/\alpha)$. Define

$$L_\phi = \bar{X}_\phi^c - \sqrt{2s_\phi^2 \ell / n} - \frac{14b_\phi \ell}{3(n-1)} - E_{\phi,h}^{\text{all}} - \tau_\phi. \quad (43)$$

In the implementation the empirical mean is enclosed downward and the variance and every subtracted quantity upward. The upper payoff endpoint uses the same terms with reversed signs. The factor F counts both tails when both are reported.

THEOREM 9—Policy-specific continuous-time certificate: *Condition on the training and validation data, and suppose the specified final simulation paths are independent draws from the declared innovation law. Under the arithmetic contract and transfer assumptions above, with probability at least $1 - \alpha$, simultaneously for all declared policies and initial states,*

$$J_y(a^\phi) - J_y(m_0) \geq L_\phi, \quad 0 \leq V^*(y) - J_y(a^\phi) \leq G_d - L_\phi. \quad (44)$$

The optimum V^ still ranges over all adapted feasible controls in the original continuous economy, not merely the networks, schedules, or sampled controllers used in the experiment. Common shocks may be reused across policies; independence across those policies is not required. The guarantee is pointwise in each declared initial state and conditional on training, rather than uniform on an uncountable state domain.*

PROOF: Apply the empirical Bernstein inequality of [Maurer and Pontil \(2009, Theorem 4\)](#) to $(X_{\phi,h}^c + b_\phi)/(2b_\phi)$ and use a union bound over the declared one-sided statements. Rescaling gives the first two error terms in (43). The clipping expectation bound and Proposition 8 then give the first inequality in (44). Subtract it from the deterministic anchor bound (33). Feasibility yields nonnegative regret. An observed

negative upper regret endpoint would indicate failure of the confidence event or of an implementation assumption, not a negative optimality gap. *Q.E.D.*

Unlike $G_d + D_{d,\epsilon'}$, the new endpoint depends on the deployed policy through its paired outcomes and sample variance. It can distinguish seeds and unsuccessful policies in the same architectural tube. When $L_\phi > 0$, the post-training guarantee is tighter than the analytical schedule’s guarantee. This is a statement about the verified policy, not a deterministic promise that every training run improves welfare. The concentration inequality, the identity that transfers a lower improvement bound into a regret bound, and conservative policy improvement are established ideas (Kakade and Langford, 2002, Pirotta et al., 2013, Scherrer et al., 2015). The contribution here is their operational connection to the continuous capital economy through (40)–(42), without a state-covering certificate or a global bound on the trained network’s derivatives.

8. BELLMAN EVALUATION WITH COSTATE TARGETS

The post-training certificate applies to any admissible candidate; it does not select a numerical method in advance. We therefore change the evaluation experiment as well as the verification step. The purpose is to determine whether a separately trained continuation value and feasible actor produce economically useful state dependence, rather than merely remain inside a safe architectural interval.

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\hat{V}^{(j)}(t, y)$ and its pathwise derivative $\hat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\mathbb{E} \left[.05 \{v_\theta - \hat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \hat{p}^{(j)})\|_d^2 \right]. \quad (45)$$

The factor d puts normalized marginal values on an order-one scale. The critic has the same two hidden tanh layers and hard terminal component as the R10 implementation. Holding its new derivative fixed, the actor then maximizes

$$\hat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (46)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. Five critic steps and five actor steps are performed for each fresh rollout batch. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

8.1. Comparison and selection

The primary comparisons include direct policy optimization through the same simulator and an affine state-dependent actor followed by tanh and the same feasibility guard. Direct policy optimization has the same nonlinear actor architecture but no learned critic; the affine method exploits a simpler representation. They receive the same mixture of training states, the same seeds, the same rollout length, and the same number of simulation-state visits. Their optimizer-step counts differ by design and are reported, as are wall time and deployment cost. Equality of simulator work is not described as equality of total computation.

Every twentieth iteration is saved. Each method chooses the checkpoint with the highest return on its own fixed validation bank before any final-test shocks are examined. The archive includes all six checkpoints, not only the selected one. Longer

direct-policy and affine runs expose whether a short baseline budget is responsible for a comparison. Cost-accuracy frontiers use the saved checkpoints and the same final statistical accounting. The method is not declared superior merely because one direct greedification of an inaccurate critic performs poorly.

Training and validation use finite Euler rollouts. Verification instead uses the specified sampled controller, an independent fine-grid paired statistic, and Proposition 8. An empirical rollout improvement is therefore not substituted for the continuous-time conclusion. Conversely, no uniform HJB residual is assumed merely because a rollout or costate loss is small.

9. THE ECONOMIC VALUE OF STATE-DEPENDENT CONSUMPTION

The quantitative question is whether conditioning consumption on the evolving, densely coupled capital state improves upon a common feasible consumption plan enough to justify the computational intervention. This question is posed in the original benchmark economy; the exercise is not a calibration to household or firm data. The relevant comparison is the schedule itself, not just another neural implementation. All reported gains are differences in the original discounted utility criterion.

9.1. *Design and primary evidence*

The final design uses ten independent training seeds in each of dimensions ten, twenty, and fifty. Each seed supplies an NBO actor, a direct-policy actor, and an affine actor. The parameter vector and dense matrix-generation rule are unchanged from R10. The primary candidate architecture has width 32, two hidden layers, radius .1, and 120 outer iterations. Each selected policy is evaluated from $y = 0$ using 8,192 fresh paths and 1,024 time cells. A family allocation of 2,000 one-sided statements at total probability .05 covers the primary comparisons and all declared sensitivity rows. The final noise seed differs from the exploratory bank used during development. This is a design fixed before the final remote computation, not preregistration before all research.

Of the 30 primary NBO policies, 30 have a strictly positive simultaneous lower improvement endpoint. The lower endpoints range from 0.000716 to 0.001193 in

TABLE II

PRIMARY CONTINUOUS-ECONOMY COMPARISON. TEN TRAINING SEEDS PER ROW; ALL POLICIES START AT $y = 0$. GAIN IS IN DISCOUNTED UTILITY UNITS. LOWER ENDPOINTS ARE ROUNDED DOWN AND UPPER ENDPOINTS UP.

d	Method	Mean gain	Seed s.d.	Min. L_ϕ	Max. regret
10	NBO	0.001963	0.000003	0.001183	0.017564
10	Direct policy	0.001886	0.000004	0.001105	0.017642
10	Affine	0.001190	0.000008	0.000408	0.018339
20	NBO	0.001582	0.000003	0.000716	0.024810
20	Direct policy	0.001498	0.000004	0.000629	0.024896
20	Affine	0.001020	0.000006	0.000151	0.025374
50	NBO	0.001752	0.000001	0.000868	0.026672
50	Direct policy	0.001645	0.000004	0.000757	0.026783
50	Affine	0.001025	0.000009	0.000130	0.027410

discounted utility units. The number meeting the declared improvement target .0005 is 30. The largest policy-specific NBO regret upper endpoint is 0.026672. These are results for the stated implementation, initial state, and conditional sampling contract. The archive contains 230 two-sided evaluation records and 0 logged training failures; an unmet gain target is recorded separately from a training exception.

The table reports the range across fitted seeds, including the least favourable seed, rather than a confidence interval obtained after averaging fitted actors within each path. The family-wise endpoints describe simulation uncertainty conditional on each fitted object. The accompanying seed standard deviations describe algorithmic variability; they are not asserted to estimate the distribution of every possible training run. The raw seed-by-path outcomes permit both quantities to be recomputed.

9.2. Accuracy, cost, and the role of training

The feasible schedule has zero training cost. Its deterministic regret bound is printed beside the policy-specific endpoints. The reported verification cost includes fine-grid simulation and enclosure construction; it is not hidden in a training-only runtime. Once a policy has been verified, deployment requires the stored actor and

TABLE III

RECORDED CPU COSTS AND THE DECLARED IMPROVEMENT TARGET. VERIFICATION INCLUDES SETUP, SIMULATION, AND ARITHMETIC. DECISION TIME INCLUDES A BATCH OF 256 ACTION AND INTERNAL-DRIFT UPDATES. THE LAST COLUMN COUNTS $L_\phi \geq .0005$ AMONG TEN SEEDS.

d	Method	Train (s)	Verify (s)	Decision (ms)	Target / 10
10	NBO	5.53	17.53	0.447	10
10	Direct policy	3.77	14.56	0.446	10
10	Affine	2.77	9.49	0.259	0
20	NBO	5.91	25.64	0.521	10
20	Direct policy	4.16	22.44	0.523	10
20	Affine	3.20	17.16	0.334	0
50	NBO	7.33	50.32	0.755	10
50	Direct policy	5.37	47.32	0.751	10
50	Affine	4.39	41.76	0.566	0

TABLE IV

ANALYTICAL COMPARATORS BEFORE POLICY-SPECIFIC IMPROVEMENT. THE LAST TWO COLUMNS ARE EXTERNALLY FINANCED FLOW-EQUIVALENT PERCENTAGES, NOT EMPIRICAL POPULATION WELFARE EFFECTS.

d	G_d	$G_d + D_{d,1}$	Anchor (%)	Tube (%)
10	0.018747	0.057430	1.931	6.034
20	0.025526	0.067244	2.639	7.101
50	0.027540	0.070081	2.850	7.411

an internal state update, but no critic optimization or global action search. Machine timing is a property of the recorded CPU run, not a dimension-free complexity theorem.

The archived R10 actors had negative actor-minus-schedule payoff differences in dimensions twenty and fifty. Those outcomes are retained. They are not relabelled as results of the new rollout-costate implementation. The current comparison changes policy evaluation and specifies sampled deployment; the new continuous-time transfer bound accounts for the latter. It does not isolate every source of the differ-

ence from R10, and it does not turn a method-specific experiment into a universal ranking of actor-critic and direct methods.

TABLE V
CHECKPOINT FRONTIER IN DIMENSION FIFTY, SEED 11. EACH CHECKPOINT HAS 2,048 PATHS AND 512 CELLS; THE EXPLICIT TRANSFER BOUND IS RETAINED. A NEGATIVE LOWER ENDPOINT IS NOT REPLACED BY A PASS.

d	Method	Iteration	Mean gain	L_ϕ	Regret upper
50	Direct policy	20	0.000970	-0.001564	0.029103
50	Direct policy	40	0.001420	-0.001116	0.028655
50	Direct policy	80	0.001611	-0.000928	0.028467
50	Direct policy	120	0.001645	-0.000894	0.028434
50	Affine	20	0.000507	-0.002029	0.029568
50	Affine	40	0.000614	-0.001919	0.029459
50	Affine	80	0.000836	-0.001697	0.029236
50	Affine	120	0.001012	-0.001521	0.029061
50	NBO	20	0.001717	-0.000823	0.028362
50	NBO	40	0.001743	-0.000797	0.028337
50	NBO	80	0.001747	-0.000793	0.028332
50	NBO	120	0.001750	-0.000790	0.028330

Checkpoint frontiers and extended-budget baselines make improvement per unit of work visible. A larger actor class has to earn its extra cost relative to the affine policy and the direct-policy method. These comparisons are retained even when a simpler method is competitive. All final policies, losses, failures, selected iterations, and peak memory records appear in the replication archive.

9.3. Economic normalization and stress

Let $A = \int_0^T e^{-\rho t} dt$. We report $100\{\exp(\Delta/A) - 1\}$ as a flow-consumption-equivalent normalization of a utility difference. It describes an externally financed log-flow supplement holding the reference resource allocation and other payoff terms fixed. It is not a self-financing change in consumption and is not an estimated welfare percentage for a calibrated population. A positive lower gain also bounds

an admissible additional implementation charge in utility units. This interpretation makes both the magnitude of the learned benefit and its practical limits visible.

TABLE VI
FROZEN-CORRECTION RADIUS COMPARISON, SEED 11. RADIUS .10 USES 8,192 PATHS; THE RESCALED RADII USE 4,096. ALL USE 1,024 CELLS AND THEIR OWN SIMULTANEOUS ERROR ACCOUNT.

d	Radius	Mean gain	L_ϕ	Regret upper
10	0.05	0.001431	0.000796	0.017951
10	0.10	0.001961	0.001185	0.017562
10	0.15	0.001649	-0.000482	0.019228
20	0.05	0.001173	0.000454	0.025071
20	0.10	0.001580	0.000718	0.024807
20	0.15	0.001274	-0.000972	0.026498
50	0.05	0.001268	0.000529	0.027011
50	0.10	0.001751	0.000868	0.026671
50	0.15	0.001488	-0.000781	0.028320

The validation-tuned critic-greedy comparisons, including the explicitly projected direct-full hybrid, appear in the supplement.

The supplement reports radii .05, .10, and .15; the first and third rescale a frozen fitted correction and are not represented as retraining at those radii. Widths 16 and 64 are separately trained. Initial-state tests include shifted means, greater dispersion, and a combined shift–dispersion stress. The exact initial vectors, action saturation, and fraction of internal states outside the training box are saved. Global feasibility alone is not evidence of useful extrapolation. The statistical statements remain pointwise at these declared states, not uniform over every state between them.

9.4. Connection to the other economic applications

Recursive utility, endogenous preference formation, temporal selves, and dynamic games remain applications of the NBO evaluation–improvement distinction, not consequences of the additive-noise capital theorem. Their full model equations, original calculations, counterexamples, tables, and proofs have been relocated with-

out deletion to the supplement. The recursive problem requires its utility-domain and comparison conditions. The preference certificates refer to their specified nodal and continuous-action objects, with continuous-state, time, and stopping transfer still separately accounted for. The strategic results bound deviations in the saved finite dynamic games. The sophisticated-self problem uses its own equilibrium deviation condition.

This organization preserves the original research program while placing one complete computation and its new policy-specific guarantee at the centre of the article. Neither the broader title nor the retention of those applications enlarges the scope of Theorem 9. The final source records a byte-verified relocation map and exact pre-revision roots, so the change in exposition can be distinguished from a change in scientific content.

10. CONCLUSION

Neural Bellman Operators connect policy evaluation, feasible improvement, and independent economic error accounting. The capital application now distinguishes an architectural performance envelope from a policy-specific post-training certificate. A paired payoff identity, explicit diffusion-transfer estimate, and finite-family sampling bound identify the value of a deployed neural correction relative to an analytical schedule. The economic comparison remains the original continuous problem and its full adapted policy class.

The rollout-costate implementation evaluates the current policy before improving its Hamiltonian. Direct-policy, affine, and tuned critic-greedy comparisons expose the cost and accuracy of that implementation without presuming its superiority. A positive verified gain is evidence about a particular deployed policy; it is neither an optimizer convergence theorem nor a claim about every initialization. The reported improvement is a quantitative result in the stated stylized economy, not an empirical calibration.

Recursive utility, endogenous preferences, sophisticated temporal selves, dynamic games, viscosity selection, and stochastic traces remain within the paper's economic program. Their full equations and evidence are retained in the supplement, with their own domains and verification hypotheses. Extending the new

diffusion-transfer calculation to other observation rules, state-dependent diffusion, recursive utility, or stopping boundaries requires those additional error terms to be verified. The present theorem does not silently supply them.

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